

ATmega16A Based GPS Receiver

ASHUTOSH M. BHATT

A global positioning system (GPS) receiver is used to get precise geographical location by receiving information from satellites. It not only gives information about location but also information like time, date, height and speed. It is so useful that most smartphones are embedded with it.

GPS receivers have many applications in aircraft, ships, sea vessels and the like for navigation purposes. Smartphones with maps (like Google Maps) find routes to a specific destination such as a restaurant, hospital or hotel. With the help of a GPS, a

target camp can be located and a missile launched to destroy it.

The GPS receiver module includes an antenna and a built-in processor. The built-in processor calculates the required information from the signals' output serially in form of strings using NMEA (expanded as National Marine Electronics Association) 0183 protocol. These strings are received serially by the host computer for display or by host processor/controller to take any decision or action.

NEMA protocol includes standard messages given by the GPS receiver. Some standard message outputs from NMEA are GGA, ZDA, VTC, RMC, GSA and GSV. The string starts with \$GPRMC tag and gives time, latitude, longitude, etc.

The project here demonstrates how to get location (latitude and longitude), time, date, speed and course-angle information using a GPS receiver. It uses 8-bit AVR microcontroller (MCU) ATmega16A



(ATMega32 was used by the author) to get data from the GPS receiver and display it on an LCD. All this information is combined in a single string that starts with \$GPRMC tag.

RMC stands for Recommended Minimum GNSS data. GPRMC string format is explained below, and shown in Fig. 2.

GPRMC string format

The complete string is of 65 to 70 characters including start tag. The time, that is, Greenwich Mean Time (GMT) or Coordinated Universal Time (UTC) immediately follows start tag. It can be in 1ms, 10ms or 100ms resolution. Then there is location information that includes latitude and longitude with direction.

Next is the speed of the GPS antenna, which is given in knots. Course angle is the angle of the GPS antenna with respect to true North in counter-clockwise direction. The date is given at the end. It is also with respect to GMT/UTC.

Therefore in this single string we get all the information. The AVR MCU extracts this string from GPS output and further extracts all required details and displays these on the LCD in proper format. Author's prototype is shown in Fig. 1. GPS receiver used for this prototype is a marine GPS receiver MR-600 and a sunrom GPS receiver (1216) with an active antenna.

Circuit and working

As shown in Fig. 3, the circuit is built with AVR MCU ATmega16A (IC1), RS232-level converter chip MAX232 (IC2) and a 16 × 4 LCD panel (LCD1). The GPS module has three ter-

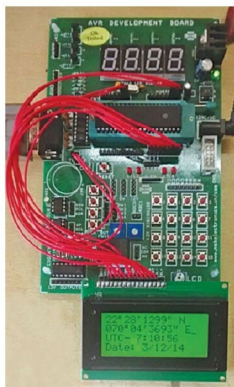


Fig. 1: Author's prototype

PARTS LIST

Semiconductors:

- IC1 - ATmega16A MCU
- IC2 - MAX232 driver

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 10-kilo-ohm
- R2 - 100-ohm
- PR1 - 1-kilo-ohm preset

Capacitors:

- C1-C4, C9 - 10 μ F, 16V electrolytic
- C5, C6 - 22pF ceramic disk
- C7, C8 - 0.1 μ F ceramic disk

Miscellaneous:

- XTAL1 - 8MHz crystal oscillator
- LCD1 - 16×4 alphanumeric display
- S1 - Tactile switch
- CON1 - 2-pin connector
- CON2 - 4-pin connector

\$GPRMC, 121040.000,A, 3354.1133,S, 15110.5941,E, 0.40, 349.50, 080211,,A*73

start tag	time(GMT)	latitude	longitude	date
		latitude direction	longitude direction	
	A - means data valid		COG - course over ground	
	V - means Navigation receiver warning			
			SOG - speed over ground (of GPS antenna)	

Fig. 2: GPRMC string format